EPL Match Outcome Predictor

Checkpoint 2: Model Extensions, Evidence and Live Testing

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Where we are. A from-scratch Poisson model, with hand-derived and numerically verified gradients, is fitted and evaluated by walk-forward validation on 4,250 matches. We added five extensions and tested each on three separate windows. The honest summary is short.

- Plain Poisson loses to the bookmakers by 0.020 log loss on 2023/24 to 2025/26.
- The best extension, which uses shots on target from earlier matches, shrinks that gap to about 0.012. Pooled over three windows its gain is 0.003 ± 0.0017, which is suggestive but not conclusive.
- The promoted-team offsets, which I adopted as a judgment call after one good window, pool to about zero. Their evidence is weak.
- Three ideas (a Dixon–Coles term, team-specific home advantage, closed-doors matches) did not help predictions, but two of them taught us something about the data.
- The live test is running: 10 matches of 23 to 25 October are frozen for three model versions.

1 What Phase 2 set out to do

Checkpoint 1 fixed the problem, audited the data and set the bar: the bookmakers' average odds score a log loss of 0.965 on 2023/24 to 2025/26. Phase 2 had three jobs.

1. Implement the Poisson model by hand and prove that the implementation is correct.
2. Add the ideas from the roadmap, one at a time, and keep only what the evidence supports.
3. Set up a live test that cannot be gamed after the fact.

Model A (a multinomial logistic baseline) from the roadmap was not built. The Poisson family gave a more informative sequence of experiments, and Model A remains optional.

2 How evidence is gathered

2.1 Three windows, one decision rule

Window	Dates	Matches	Role
Early	Jul 2018 – Jun 2021	1,140	Independent check. No model was chosen or tuned on it.
Validation	Jul 2021 – Jun 2023	760	All model choices and hyperparameter tuning.
Test	Jul 2023 – Jun 2026	1,140	Looked at for many variants, so treated as indicative.

Within each window, predictions are made week by week: fit on all matches strictly before that week, predict the week, move on. Nothing from the future is used, and a unit test checks this directly by corrupting every result after a cut-off date and confirming that earlier predictions do not change. The decision rule is that *validation decides*. The early window is a second opinion that never influenced a choice. The test window is secondary because we have looked at it repeatedly.

2.2 Paired differences and their uncertainty

To compare model A with model B on the same N matches, take the per-match log-loss difference

$$d_i = \ell_i^A - \ell_i^B, \qquad \ell_i^M = -\log p_i^{r_i}(M).$$

Its mean and standard error are

$$\bar{d} = \frac{1}{N}\sum_i d_i, \qquad \mathrm{SE} = \frac{s_d}{\sqrt{N}},$$

where s_d is the sample standard deviation of the d_i. A negative $\bar{d}$ means A is better. Estimates from independent windows $k = 1, \ldots, K$ are pooled by inverse-variance weighting:

$$\hat{d} = \frac{\sum_k w_k \bar{d}_k}{\sum_k w_k}, \qquad w_k = \frac{1}{\mathrm{SE}_k^2}, \qquad \mathrm{SE}(\hat{d}) = \frac{1}{\sqrt{\sum_k w_k}}.$$

Two reasons the standard errors are too small. First, they treat matches as independent, but errors from the same team in the same season are correlated, so the true uncertainty about a team-level effect is larger. A promoted-team effect rests on nine team-seasons in the test window (three promoted clubs per season), not on hundreds of independent matches. Second, we have looked at the test window for many variants. A gain of 1.8 standard errors is unremarkable if enough variants are tried.

3 The Poisson model, verified

The model is the one derived in Checkpoint 1:

$$\log \lambda_i = \iota + \eta + \alpha_{h(i)} - \beta_{a(i)}, \qquad \log \mu_i = \iota + \alpha_{a(i)} - \beta_{h(i)},$$

fitted by maximising the weighted, ridge-penalised log-likelihood $\ell(\theta)$ with L-BFGS-B, using the analytic gradient with the residuals $e_i = x_i - \lambda_i$ and $f_i = y_i - \mu_i$.

3.1 Evidence that the implementation is correct

- **Gradient check.** Every analytic gradient is compared with a numerical (finite-difference) gradient in the test suite, for the base model and for each extension.
- **Stationarity.** At the fitted optimum $\sum_t \alpha_t = \sum_t \beta_t = 0$, as the identifiability argument predicts.
- **Recovery on simulated data.** Data simulated from known parameters are fitted and the parameters recovered, including γ and ρ.
- **No leakage.** Predictions for early weeks are unchanged when results after them are corrupted.

3.2 Hyperparameters and the first real result

The decay rate ξ and ridge strength κ are chosen on the validation window. The best setting is $\xi = 1.0$ per year, a half-life of $\ln 2/\xi \approx 0.69$ years, and $\kappa = 5$. On the test window plain Poisson scores a log loss of 0.9852.

How much of the bookmakers' advantage does plain Poisson capture? Against the class-frequency baseline (1.0728), the bookmakers improve by $1.0728 - 0.9650 = 0.1078$ and the model by $1.0728 - 0.9852 = 0.0876$. That is

$$\frac{0.0876}{0.1078} \approx 81\%,$$

leaving a gap of 0.0202 ± 0.0047 to the Average odds.

4 Extensions: the mathematics

Every extension below changes the likelihood in a small, local way. Two ideas recur: a covariate that shifts the log-rates with a learned coefficient, and a ridge penalty read as a prior.

4.1 Covariates on both teams' rates

Let each match carry covariate vectors $\mathbf{x}_i$ (entering $\log \lambda_i$) and $\mathbf{x}'_i$ (entering $\log \mu_i$), with one shared coefficient vector γ:

$$\log \lambda_i = \iota + \eta + \alpha_{h(i)} - \beta_{a(i)} + \mathbf{x}_i^\top \gamma$$

$$\log \mu_i = \iota + \alpha_{a(i)} - \beta_{h(i)} + \mathbf{x}_i'^\top \gamma.$$

The model is still linear in the parameters inside the log-rates, so the same residual calculation applies and the new gradient is

$$\frac{\partial \ell}{\partial \gamma} = \sum_i w_i \left(e_i \, \mathbf{x}_i + f_i \, \mathbf{x}'_i \right).$$

Sharing γ is what lets one number act on both sides of a match, for example a promoted team that is weaker both in attack and in defence.

4.2 Closed doors

Matches played without spectators, from 17 June 2020 to 16 May 2021 get a flag $c_i \in \{0, 1\}$ with $\mathbf{x}_i = c_i$ and $\mathbf{x}'_i = 0$, so the coefficient is a pure shift in home advantage.

Result. $\hat{\gamma}_{\text{cd}} \approx -0.15$, against a home advantage with fans of $\hat{\eta} \approx 0.18$. Without fans, $\hat{\eta} + \hat{\gamma}_{\text{cd}} \approx 0.03$: the home boost nearly vanished. This matches the 2020/21 anomaly found in Checkpoint 1 (away wins 40.3%, home wins 37.9%).

The term does not change prediction quality in the test window, which contains no closed-doors matches. Its job is to stop the 2020/21 matches from distorting the ordinary home-advantage estimate.

4.3 Dixon–Coles correction

The gradient of the correction term is where the earlier residual calculation pays off. Write $\tau_i = \tau(x_i, y_i)$. The four cases give

$$\frac{\partial \log \tau}{\partial \log \lambda} = \begin{cases} -\lambda\mu\rho/\tau & (0,0) \\ \lambda\rho/\tau & (0,1) \\ 0 & \text{otherwise} \end{cases} \qquad \frac{\partial \log \tau}{\partial \log \mu} = \begin{cases} -\lambda\mu\rho/\tau & (0,0) \\ \mu\rho/\tau & (1,0) \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{\partial \log \tau}{\partial \rho} = \begin{cases} -\lambda\mu/\tau & (0,0) \\ \lambda/\tau & (0,1) \\ \mu/\tau & (1,0) \\ -1/\tau & (1,1). \end{cases}$$

Defining the modified residuals

$$e_i' = e_i + \frac{\partial \log \tau_i}{\partial \log \lambda_i}, \qquad f_i' = f_i + \frac{\partial \log \tau_i}{\partial \log \mu_i},$$

every earlier gradient formula holds with e_i, f_i replaced by e_i', f_i', and $\partial \ell / \partial \rho = \sum_i w_i \, \partial \log \tau_i / \partial \rho$. With $\rho \neq 0$ the likelihood is no longer concave, so the optimiser can only promise a local maximum.

Result. $\hat{\rho} \approx -0.09$ to -0.096. A negative ρ raises the probability of 0–0 and 1–1 scorelines and so raises draw probabilities. The test log loss improves from 0.9851 (closed doors) to 0.9844 (closed doors plus Dixon–Coles), a gain of 0.0007, well inside the noise, and on validation the term was slightly worse (0.9741 for B plus Dixon–Coles against 0.9736 for B). It is not adopted.

4.4 Promoted-team offsets

A team is a *newcomer* in season s if it plays in s but not in $s-1$. Let $n_i^h, n_i^a \in \{0,1\}$ flag whether the home and away team are newcomers. We add an attack offset π_α and a defence offset π_β shared by all newcomers:

$$\tilde{\alpha}_t = \alpha_t + \pi_\alpha \, n_t, \qquad \tilde{\beta}_t = \beta_t + \pi_\beta \, n_t.$$

In the covariate notation of Section 5.1, with $\boldsymbol{\gamma} \ni (\pi_\alpha, \pi_\beta)$, the columns are

$$\mathbf{x}_i = \big(n_i^h, \, -n_i^a\big), \qquad \mathbf{x}_i' = \big(n_i^a, \, -n_i^h\big),$$

where the minus signs appear because defence enters the log-rates with a negative sign.

Reading the ridge as a prior. The penalty $\kappa(\alpha_t - m)^2$ is the negative log-density of the prior $\alpha_t \sim \mathcal{N}\big(m, \frac{1}{2\kappa}\big)$ up to a constant. Before, every team had $m = 0$. With the offsets, a newcomer's prior mean is π_α (and π_β for defence), and π is itself learned from all earlier promoted teams. This is a hierarchical prior.

Two-stage estimation. The offsets describe promoted clubs in general and should change slowly, whereas team strengths should be forgotten quickly. In the two-stage variant, stage one estimates π with its own decay rate (here none), and stage two fits everything else with the usual decay ξ while π is held fixed. The fixing is done by giving the optimiser equal lower and upper bounds for those two parameters.

Result. Two-stage offsets are $\hat{\pi}_\alpha = -0.172$ and $\hat{\pi}_\beta = -0.107$: a newcomer scores about 16% fewer goals and concedes about 11% more than its ridge-shrunk strength implies. A joint fit gives larger offsets (-0.246 and -0.200), so recent promoted sides look weaker than earlier ones.

Against the closed-doors model, the change in log loss is $+0.0008 \pm 0.0007$ (early), 0.0000 ± 0.0006 (validation) and -0.0036 ± 0.0012 (test). Pooled, this is -0.0002 ± 0.0004: no net evidence. I adopted the offsets after seeing only the test window, and the early window does not support that call. They are harmless but unproven.

4.5 Team-specific home advantage

Each team gets a deviation δ_t from the common home advantage, penalised by a separate ridge:

$$\log \lambda_i = \iota + \eta + \delta_{h(i)} + \alpha_{h(i)} - \beta_{a(i)}, \qquad \ell \;\to\; \ell - \kappa_h \sum_t \delta_t^2.$$

The gradient has the same shape as the one for η but restricted to team t's home matches:

$$\frac{\partial \ell}{\partial \delta_t} = \sum_{i:\, h(i)=t} w_i \, e_i - 2\kappa_h \, \delta_t.$$

The prior standard deviation is $1/\sqrt{2\kappa_h}$: 0.32, 0.16 and 0.09 for $\kappa_h = 5, 20, 60$. The ridge is needed because each team has only a few dozen home matches in the effective training window, so an unpenalised δ_t would mostly fit noise.

Result. The fitted spread of δ is small: standard deviation 0.051 at $\kappa_h = 20$. Newcastle comes out highest ($+0.10$) and Nott'm Forest lowest (-0.14), but the most extreme of 20 teams always looks striking. On validation the best team-specific setting ($\kappa_h = 20$, 0.9736) did not beat the common home advantage (0.9733). Not adopted.

4.6 Shots on target as evidence of team strength

Goals are rare and noisy; shots on target occur several times more often per match, so they are a less noisy sign of quality. The model uses them in two stages, using only matches before the prediction date.

Stage 0. Fit a Poisson model to shots on target (s_i^h, s_i^a) with the same structure,

$$\log \sigma_i^h = \iota_s + \eta_s + a_{h(i)} - b_{a(i)}, \qquad \log \sigma_i^a = \iota_s + a_{a(i)} - b_{h(i)},$$

with its own decay rate ξ_s and ridge strength $\kappa_s = 5$. This gives shot-based strengths a_t (attack) and b_t (defence).

Stage 1. Feed them to the goal model through one learned coefficient c:

$$\log \lambda_i = \iota + \eta + \alpha_{h(i)} - \beta_{a(i)} + c\, z_i, \qquad z_i = a_{h(i)} - b_{a(i)}$$

$$\log \mu_i = \iota + \alpha_{a(i)} - \beta_{h(i)} + c\, z_i', \qquad z_i' = a_{a(i)} - b_{h(i)}.$$

This is the covariate form of Section 5.1 with $\mathbf{x}_i = z_i$ and $\mathbf{x}'_i = z'_i$, so

$$\frac{\partial \ell}{\partial c} = \sum_i w_i \big(e_i\, z_i + f_i\, z'_i\big).$$

The penalty $\kappa\, \alpha_t^2$ now reads $\kappa\big(\text{effective attack} - c\, a_t\big)^2$, where the effective attack is $c\, a_t + \alpha_t$. Each team's prior mean strength comes from its shots instead of being zero. A team that scores more than its shots justify is treated as partly lucky.

Result. The fitted coefficient is $\hat{c} = 1.35, 1.55, 1.61$ for $\xi_s = 0.3, 0.6, 1.0$.

A prediction of mine that was wrong. I expected c between 0.5 and 1. It is above 1. One possible reason is that better teams also shoot from better positions, so shots on target understate the real quality gap. I have not tested that explanation.

5 Results

5.1 Test window, all variants

Model	Val LL	Test LL	Gap to Avg odds
Poisson (B)	0.9736	0.9852	$+0.0202 \pm 0.0047$
B + closed doors	0.9733	0.9851	$+0.0201$
Closed doors + Dixon–Coles (C)	0.9738	0.9844	$+0.0194$
Promoted offsets, joint fit	0.9746	0.9789	$+0.0139$
Working: closed doors + promoted, two-stage	0.9733	0.9815	$+0.0165 \pm 0.0046$
Working + team home advantage ($\kappa_h = 20$)	0.9736	0.9805	$+0.0155$
Working + shots ($\xi_s = 0.6$)	0.9747	0.9772	$+0.0122$
Working + shots ($\xi_s = 1.0$)	0.9740	0.9770	$+0.0119 \pm 0.0046$
Average odds, margin removed	–	0.9650	–

Gaps without a standard error are differences of the printed log losses to 0.9650. The Average-odds bookmaker is right on 54.2% of matches (Checkpoint 1). Differences among our models are a few thousandths of log loss.

5.2 The same ideas on the early window

Model, 2018 – 2021	Log loss	Gap to B365 odds
Poisson (B)	0.9625	+0.0048
B + closed doors	0.9628	+0.0051
Closed doors + Dixon–Coles (C)	0.9653	+0.0076
Promoted offsets, joint fit	0.9641	+0.0064
Working	0.9635	$+0.0058 \pm 0.0052$
Working + team home advantage	0.9634	+0.0057
Working + shots ($\xi_s = 0.6$)	0.9593	$+0.0016 \pm 0.0046$
Working + shots ($\xi_s = 1.0$)	0.9601	$+0.0024 \pm 0.0046$

The decay and ridge settings were fixed before this window was scored (shots models use $\xi = 1.5$, $\kappa = 5$), and training history is only three seasons at the start of the window.

5.3 Pooling the evidence



Figure 1: Change in log loss against the reference model in each window, with 95% intervals (±1.96 SE) and the inverse-variance pooled estimate. Left of zero is better. Shots on target: reference is the working model. Promoted offsets: reference is the closed-doors model.

Shots on target. Pooled gain -0.0031 ± 0.0017 for $\xi_s = 1.0$ and -0.0032 ± 0.0017 for $\xi_s = 0.6$, about 1.9 standard errors. The signs agree in the early and test windows. Validation, the smallest window, is slightly the other way, but its interval (± 0.008) is wide enough to contain the effect seen elsewhere.
Promoted offsets. Pooled -0.0002 ± 0.0004, no net evidence.

5.4 The gap to the bookmakers, season by season

Earlier in this project I compared whole windows and described the gap as widening from about 0.005 (2018 to 2021) to about 0.019 (2023 to 2026). A season-by-season run (`src/gap_by_season.py`, same walk-forward procedure, settings of the three live models) shows that description was too simple. The gap to the B365 odds, in log loss, with SE ≈ 0.008 per season:

Season	Goals/match	v0 Poisson	v1 working	v2 shots
2018/19	2.82	+0.0064	+0.0081	−0.0043
2019/20	2.72	+0.0011	+0.0041	+0.0039
2020/21	2.69	+0.0067	+0.0053	+0.0076
2021/22	2.82	+0.0203	+0.0211	+0.0204
2022/23	2.85	+0.0235	+0.0221	+0.0243
2023/24	3.28	+0.0302	+0.0271	+0.0202
2024/25	2.93	+0.0177	+0.0083	+0.0056
2025/26	2.75	+0.0093	+0.0107	+0.0065
2026/27 (70 matches)	2.90	+0.0086	+0.0156	+0.0075

What the season split shows.

- The gap is a *bump*, not a trend. It was about 0.005 in 2018 to 2021, about 0.02 to 0.03 in 2021/22 to 2023/24, and has come back to about 0.01 in the two most recent full seasons (plus 70 matches of 2026/27).
- The test window's 0.0202 is mostly this bump: 2023/24 alone contributes 0.030.
- Pinnacle's odds give the same picture to within about 0.002 in every season, so a sharper bookmaker does not change the story.
- The shots model is the best or tied for best in every season except 2019/20, 2020/21 and 2022/23, and its gap in 2024/25 to 2026/27 is 0.006 to 0.008, which is within about one standard error of zero per season.

Is it goal-rate drift? Only partly. 2023/24 had an unusually high 3.28 goals per match, and the plain model's gap is largest there. But in 2021/22 and 2022/23 goals per match were ordinary (2.82 and 2.85) and the gap was already 0.02 for *every* model, including the shots model. Across seasons the correlation between the gap and the change in goals per match is $+0.42$ for v0, $+0.68$ for v1 and $+0.39$ for v2 (nine points, so weak evidence either way).

Is it a bias in average outcomes? No. In 2022/23 home wins were 48.4% against a plain-model prediction of 43.9%, which looks like a miss, but the bookmakers' average was 43.7%, the same miss. Across seasons the bookmakers' average probabilities are no better calibrated than the model's. The bookmakers' advantage in 2021/22 to 2023/24 therefore came from separating *individual matches* better, not from knowing the league-wide mean. This points to match-specific information (team news, form, lineups) or to team strengths changing faster than a decay rate of $\xi = 1$ captures, but this data cannot say which.

What this changes. The live test now has a clearer question: do the models sit near 0.01 behind the bookmakers, as in the latest two seasons, or near 0.02, as in the test window? The 70 matches of 2026/27 already lie near the lower value, but 70 matches have a standard error of about 0.03 and say nothing yet.

6 Live testing

6.1 Design

A backtest, however carefully built, is run by someone who has seen the outcomes. The live test removes that: predictions are written to a log *before* kickoff and never changed.

- Each prediction row is keyed by match and model version. The log is append-only: a row that is already frozen is skipped on a rerun, never overwritten.
- Fixtures that have already started are skipped, and the kickoff time is compared with UK time.
- Three versions run side by side on the same matches: `v0-poisson` (plain Poisson), `v1-working` (closed doors and promoted offsets) and `v2-shots` (v1 plus shots on target).
- The script refuses to predict if the shot data for recent matches is out of date.
- A scoring command pairs each frozen prediction with its result and compares log loss, Brier score and accuracy with the bookmakers.

6.2 Round 1

Ten matches from 23 to 25 October 2026 were frozen on 20 October for all three versions (30 rows). The fixture list comes from the Premier League's published schedule, because the data site's fixtures file still listed the round that had just been played. Each row records the model version, the number of training matches (4,250) and the last training date (19 October 2026). The probabilities are in the log (`live/predictions_log.csv`) and are not reproduced here.

6.3 How much live data is enough?

Suppose a pair of predictors differ by Δ in mean log loss, and the per-match differences have standard deviation s. The difference reaches two standard errors when

$$N = \left(\frac{2s}{\Delta}\right)^2.$$

From the test-window standard errors, $s \approx 0.155$ for model against bookmaker (the standard error 0.0046 at $N = 1{,}140$ gives $s = 0.0046\sqrt{1140}$) and $s \approx 0.084$ for shots against working.

$$\Delta = 0.0165 \text{ (model vs bookmaker)}: \qquad N = \left(\frac{2 \times 0.155}{0.0165}\right)^2 \approx 350$$

$$\Delta = 0.0032 \text{ (shots vs working)}: \qquad N = \left(\frac{2 \times 0.084}{0.0032}\right)^2 \approx 2{,}800$$

What this means. The gap to the bookmakers can be confirmed within about one season (380 matches). Whether shots really help by about 0.003 cannot: it would take roughly seven seasons of live data. The live test settles the first question and can only add a little to the second.

7 Limitations

- **Repeated looks.** The test window was scored for many variants. Its gains should be read as indicative, and the early window and live data are the independent checks.
- **Standard errors.** They assume independent matches and ignore team-level correlation. They are lower bounds on the real uncertainty.
- **Few team-seasons.** Promoted-team effects rest on 6 team-seasons in validation

and 9 in the test window.

- **Missing information.** No injuries, lineups, managerial changes, fixture congestion, or market information. These are the likely reason bookmakers keep an edge.
- **Benchmark choice.** The Average odds are missing before 2019/20, so the early window is compared with B365, a slightly weaker benchmark. Pinnacle odds were checked per season and agree with B365 to about 0.002; true closing odds are not used.
- **First round.** Ten matches cannot separate any two models. They check the pipeline, not the models.

8 Bugs, errors and corrections

What went wrong	Fix
Downloads from the data site failed from the Mac until the server address was pinned.	Use `curl -resolve` with the working address, keeping the apex URL.
Several tests failed on tolerances that were too tight: parameter recovery of team home advantages (max error 0.05) and a huge-ridge test that reduces to the plain model.	Relaxed to the real sampling error (0.1) and used $\kappa_h = 10^4$ with absolute tolerances of 2×10^{-3} and 5×10^{-3}, since 10^7 makes the optimisation stiff.
A live-versus-backtest consistency test failed.	Test bug: the backtest refits at the start of each calendar week and my window crossed a week boundary. After aligning the window to one calendar week, live and backtest predictions agree.
The fixtures file listed only matches already played, so nothing could be predicted.	The kickoff guard correctly refused. A hand-made file from the published schedule was used.
The first batch of live predictions was timestamped in the Mac's local time (IST), not UK time.	Verified the training cut-off (4,250 matches, last 19 October 2026); later batches carry a UK offset.
My prediction that the shots coefficient would lie between 0.5 and 1 was wrong (1.35 to 1.61).	Interpretation revised: goal rates respond more than one-for-one to shot-implied strength.
I adopted the promoted-team offsets as a judgment call after the test window alone.	The early window shows no net evidence; status downgraded to unproven.

9 Next steps

1. Keep freezing each matchweek, refreshing results and shots data afterwards (`python -m src.add_shots`), and score the log after each round.
2. Decide, once a few hundred live matches exist, whether the gap to the bookmakers is stable, and revisit shots versus working with the full evidence.
3. Optional: Model A (Elo and rolling form) as a baseline and for stacking; closing odds as a stricter benchmark; a test of faster team-strength decay in seasons where the gap is large.
4. Phase 4: final report with derivations, a calibration plot, and an honest account of which gains survived.

Numbers in this report come from the outputs of `src/compare_models.py` and `src/early_check.py` (4,250 matches, data to 19 October 2026).